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Lithium-rich borate coated $LiMn_{0.6}Fe_{0.4}PO_4$ cathodes for high-performance and stable lithium-ion batteries

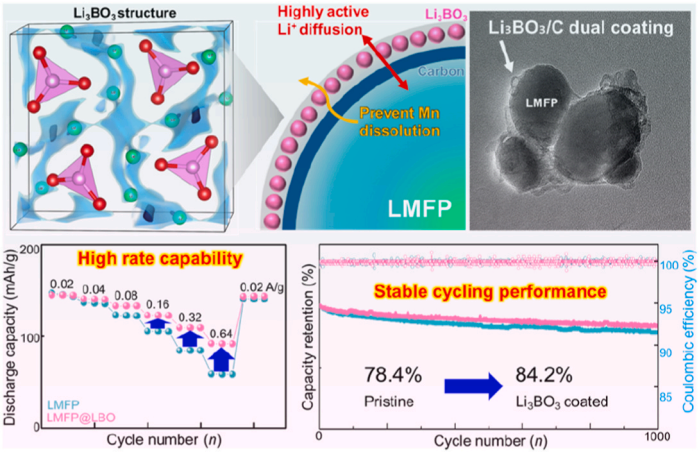
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HIGHLIGHTS

- Lithium-rich amorphous Li_3BO_3 selected via soft-BV modeling with low Li^+ barrier.
- LMFP@LBO delivers 94.04 mAh g^{-1} at 0.64 A g^{-1} with enhanced rate capability.
- Coating suppresses Mn dissolution and stabilizes electrode–electrolyte interf aces.
- Superior cycling stability: 84.2 % capacity retention after 1000 cycles at 0.16 A g^{-1}.

GRAPHICAL ABSTRACT



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ABSTRACT

The development of high-energy-density and long-life lithium-ion batteries necessitates the design of cathode materials with enhanced electrochemical and structural stability. In this work, we demonstrate a lithium-rich Li_3BO_3 surface coating as an effective strategy to improve the performance of $LiMn_{0.6}Fe_{0.4}PO_4$ cathodes. Guided by diffusion barrier calculations, Li_3BO_3 is selected for its favorable lithium-ion conductivity and low migration barrier of approximately 0.321 eV. Structural and spectroscopic analyses confirm the formation of a uniform Li_3BO_3 layer that preserves the integrity of the carbon coating while introducing hierarchical porosity. The coating significantly enhances charge-transfer kinetics and lithium-ion diffusion, resulting in superior rate capability with a discharge capacity of 94.04 mAh/g at 0.64 A/g and improved long-term cycling stability with 84.2 % capacity retention after 1000 cycles at 0.16 A/g. Furthermore, post-cycling characterizations reveal that

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the coating effectively suppresses manganese dissolution, mitigates surface degradation, and stabilizes the electrode–electrolyte interface. This study highlights the potential of Li_3BO_3 as a multifunctional coating for high-performance LMFP-based cathodes and offers a scalable strategy for advanced lithium-ion battery applications.

1. Introduction

With increasing demand for sustainable energy systems, efforts to transition away from conventional fossil fuel-based power sources toward alternative energy technologies have intensified. Among these, lithium-ion batteries (LIBs) have rapidly become prevalent in various fields such as electric vehicle (EV), smartphones, laptops, and energy storage system (ESS) due to their lightweight nature, high energy density, excellent voltage characteristics, and efficient charge-discharge cycles [1–4]. As dependence on portable electronics and electric vehicles grows, LIBs play an increasingly crucial role in our daily lives. Cathode materials, significantly impacting battery cost and performance, are a critical component of LIBs, making the enhancement of their energy density, reduction in production costs, and improved safety urgent objectives within the battery industry [5].

Recently, $LiFePO_4$ (LFP) has gained substantial attention in battery research communities due to its lower manufacturing cost and remarkable thermal stability. Structurally, LFP comprises FeO_6 octahedra and LiO_6 octahedra interconnected by PO_4 tetrahedra, providing a highly stable crystal framework during charge-discharge cycles [6–8]. However, the 1D lithium-ion diffusion channels in LFP result in negligible volumetric change (6.9 % difference between $LiFePO_4$ and $FePO_4$) but impose limitations such as a relatively low operating voltage (approximately 3.45 V vs. Li^+/Li) and inadequate high-rate charging performance.

To overcome these intrinsic limitations and improve electrochemical properties, partial substitution of Fe with Mn has led to the development of $LiMn_{0.6}Fe_{0.4}PO_4$ (LMFP) [9–12]. The introduction of Mn elevates the operating voltage from approximately 3.45 V to around 4.1 V, significantly enhancing the energy density [13]. Nevertheless, LMFP materials suffer from Jahn-Teller distortions triggered by manganese's instability upon reaction with electrolytes [14–16], inducing structural stresses and cracks, consequently diminishing long-term cycling stability. A prominent approach to address the cycling stability issues of LMFP involves the application of protective surface coatings [17,18]. The coating layer is expected to mitigate direct interaction between LMFP and electrolyte, suppress Jahn-Teller distortion, and enhance structural integrity, thus significantly improving cycling performance. Various surface coating materials have been reported to date [19–23]; however, boron-based materials have recently attracted significant attention due to their favorable ionic conductivity and suitability for mass production [17, 24–26]. Such doping and coating strategies have been widely applied to various cathode materials [27–29].

Recently, several studies have proposed different strategies to enhance the electrochemical performance of LMFP, including conductive carbon modification, cationic substitution, and surface coating design. For instance, conductive network and precursor engineering and element doping have been shown to improve charge-transfer kinetics and structural stability [30–33]. In comparison, our approach utilizing a lithium-rich Li_3BO_3 coating focuses on interfacial stabilization and suppression of Mn dissolution, achieving long-term cycling stability and superior rate performance.

In this study, we present lithium-rich borate (Li_3BO_3) as a surface coating layer for LMFP cathode. Comprehensive analysis, including electrochemical measurements, structural characterization, elemental distribution assessments, spectroscopic techniques, and Li^+ diffusion simulations, were conducted to elucidate the Li^+ diffusion pathways and corresponding energy barriers within Li_3BO_3 structure. The findings confirm that the lithium-rich Li_3BO_3 coating layer substantially improves the structural stability of LMFP particles, effectively suppresses parasitic reactions with electrolytes, and significantly enhances long-term cycling stability and electrochemical performance.

2. Experimental

2.1. Material synthesis and characterization

To synthesize Li_3BO_3, lithium hydroxide (LiOH, 98 %, Sigma-Aldrich) and boric acid (H_3BO_3, 99.5 %, Sigma-Aldrich) were dissolved in deionized water with a molar ratio of 3:1. Subsequently, 5 g of carbon-coated LMFP was added to the precursor solution, targeting Li_3BO_3 content of 1 wt% relative to the LMFP mass. The mixture was stirred using a magnetic stirrer at 400 rpm and maintained at 60 °C overnight to allow for uniform deposition and drying. The resulting powder was collected and denoted as LMFP@LBO.

Microstructural analyses of the synthesized materials were systematically conducted using scanning electron microscopy (SEM, JEOL/JSM-IT800SHL), transmission electron microscopy coupled with energy-dispersive X-ray spectroscopy (TEM-EDX, JEOL/JEM-F200), and X-ray photoelectron spectroscopy (XPS, KRATOS Analytical Ltd./AXIS SUPRA).

To analyze the crystal structure X-ray diffraction (Rigaku/UltimaIV) was conducted. Complementary analyses including FT-IR (PerkinElmer/Spectrum 3), Raman Spectroscopy (JASCO/NRS-5100), TG, DSC (TA Instruments/Discovery DSC 25, Discovery TGA 55, TMA Q400) and BET (Quantachrome/Autosorb-iQ) were performed to investigate chemical bonding, thermal behavior, and surface area characteristics.

2.2. Electrochemical tests and cell assembly

All electrochemical tests were conducted utilizing a coin-type (CR2032) cell configuration assembled in a two-electrode system. The synthesized LMFP and LMFP@LBO electrode materials were applied as the working electrodes, while metallic lithium foil served as both the reference and counter electrodes. The electrolyte employed was composed of 1.3 M $LiPF_6$ dissolved in a ternary solvent blend consisting of ethylene carbonate (EC), ethyl methyl carbonate (EMC), and diethyl carbonate (DEC) with a volumetric ratio of 3:5:2. Polypropylene separators were adopted to electrically isolate the electrodes and ensure efficient lithium-ion transport. The working electrodes were prepared by mixing the synthesized active cathode material (LMFP and LMFP@LBO), conductive carbon (Super C65, Timcal Graphite & Carbon), and polyvinylidene fluoride binder (PVDF, Kureha Corporation) in a mass ratio of 8:1:1. The resultant slurry was uniformly dispersed using N-methyl-2-pyrrolidone (NMP, Sigma-Aldrich) as the solvent. Subsequently, this slurry was evenly coated onto aluminum foil (thickness ~20 μm) using a doctor-blade method, followed by drying under vacuum conditions to remove residual solvent. After drying, the electrode-coated foils were punched into circular disks of uniform dimensions and transferred into an argon-filled glove box. The mass loading of the active material was controlled to be approximately 2.5 mg.

Coin cells were then assembled by sequentially stacking the cathode disks, separators, and lithium-metal anodes, followed by precise addition of electrolyte. Finally, the assembled cells were hermetically sealed under controlled conditions to ensure optimal electrochemical performance.

2.3. Structural analyses

The crystal structures of the synthesized samples were characterized by XRD using an Ultima IV diffractometer (Rigaku) equipped with Cu Kα radiation. To gain insights into lithium-ion migration pathways and determine the corresponding diffusion barriers, bond-valence analysis was performed via the soft-BV computational method [34,35]. The predicted lithium diffusion paths were visualized and indicated with sky-blue coloration. The crystal structures and lithium-ion diffusion channels were clearly illustrated using the VESTA3 visualization software [36].

3. Results and discussion

3.1. Selection of lithium borate (Li-B-O) composition

In selecting suitable coating material for LMFP, a set of criteria should be established first. The coating material should maintain sufficient lithium-ion conductivity, suppress Mn dissolution caused by intrinsic Jahn–Teller distortion and minimize interfacial side reactions with the electrolyte. Based on these considerations, Li-B-O emerged as a promising candidate [37–39]. Li-B-O has been previously demonstrated as a surface coating material for solid electrolytes in all-solid-state batteries to suppress interfacial degradation, prevent cathode structure collapse, alleviate electrochemical instability, and preserve Li^+ transport pathways. To apply the advantages of Li-B-O coating material for LMFP cathode, we first screened various Li-B-O polymorphs with the lowest Li^+ diffusion barrier using soft-BV calculation [27,28]. Based on the compositional phase diagram shown in Fig. 1a, six Li-B-O polymorphs in the Li_2O–B_2O_3 system were identified: Li_3BO_3, $Li_4B_2O_5$ [40], $LiBO_2$ [41], $Li_2B_4O_7$ [25,42], $Li_3B_7O_{12}$ [43], and LiB_3O_5 [44] which utilized LIB research. The Li^+ diffusion barriers for each compound were calculated and then compared. Although LiB_3O_5 exhibits the lowest energy barrier of 0.178 eV along the 1D diffusion path, it shows a significantly high barrier of 2.342 eV along the three-dimensional 3D direction. In addition, the visualized lattice structure indicated severely restricted Li^+ mobility, suggesting that this material is not suitable for fast ionic conduction. Additionally, the detailed diffusion energy barrier profiles of each Li-B-O compound are provided in Fig. S1–S5 in the Supporting Information. Consequently, Li_3BO_3, which exhibited the second-lowest Li^+ diffusion barrier of 0.321 eV, was selected as the coating material candidate. The projected crystal structure and Li^+ diffusion pathways of Li_3BO_3 are illustrated in Fig. 1b and c. Fig. 1b presents the crystal structure on the *bc*-plane, showing Li^+ migration along both the *b*-axis (0.396 eV) and *c*-axis (0.321 eV). Fig. 1c shows the structure projected on the *ab*-plane, along with the corresponding Li^+ diffusion paths.

3.2. Li_3BO_3-coated LMFP characterization

Detailed synthesis method of Li_3BO_3-coated LMFP, further referred as LMFP@LBO is available in the Experimental Section. Fig. 2a and b presents scanning electron microscopy (SEM) images along with a corresponding schematic illustration of the pristine LMFP particles. The pristine LMFP particles exhibited a relatively smooth and uniform morphology despite the presence of a carbon coating. In contrast, the SEM images and schematic illustration in Fig. 2c and d clearly demonstrate that the LMFP@LBO particles possess a rougher surface texture, indicative of the successful deposition of the Li_3BO_3 coating.

Additional SEM images along with the elemental mapping of pristine LMFP and LMFP@LBO samples using SEM equipped with energy-dispersive X-ray spectroscopy (EDX) are provided in Fig. S6–S9 of the Supporting Information. Notably, Fig. S7–S9 confirm that the original carbon coating layer remains intact after the Li_3BO_3 surface

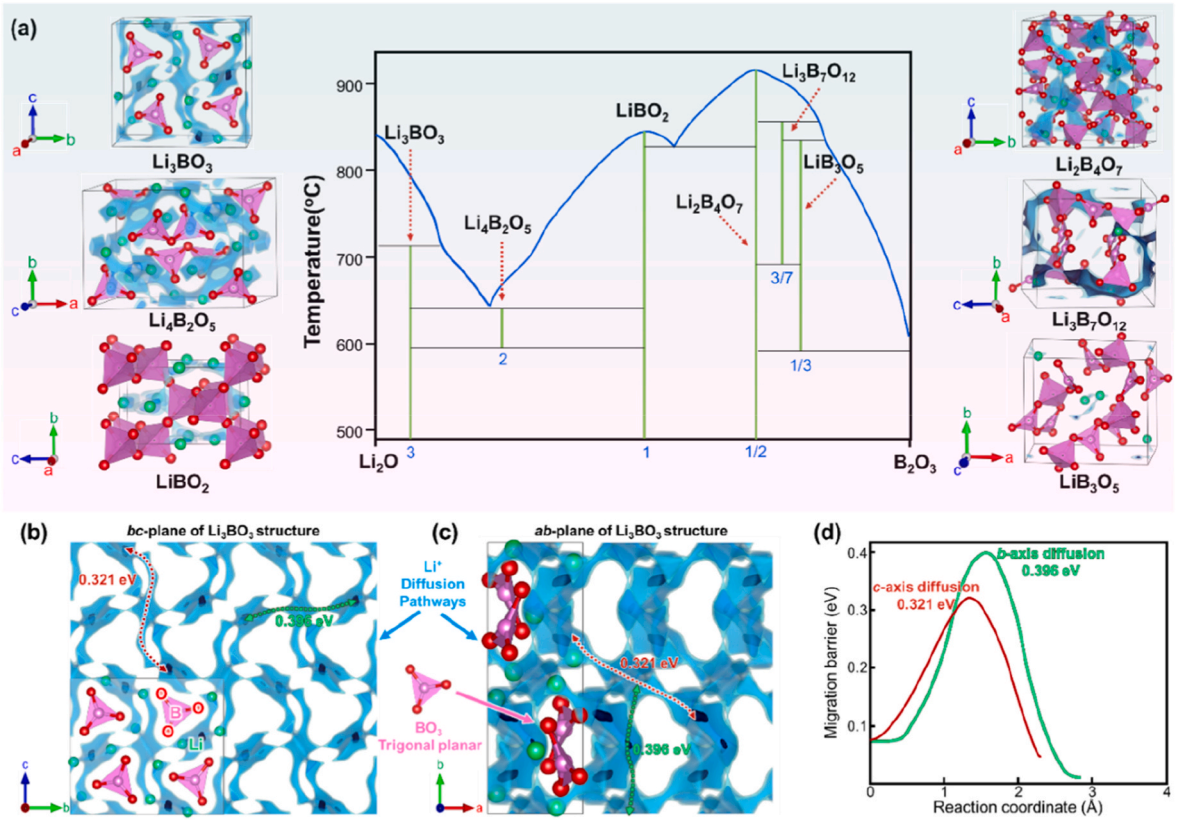


Fig. 1. (a) Compositional phase diagram of Li-B-O compounds and corresponding crystal structures with Li^+ pathways of selected compositions, (b) Crystal structure and Li^+ pathways of Li_3BO_3 on the *bc*-plane, (c) Crystal structure and lithium pathways of Li_3BO_3 on the *ab*-plane. (d) Li^+ migration barriers of Li_3BO_3.

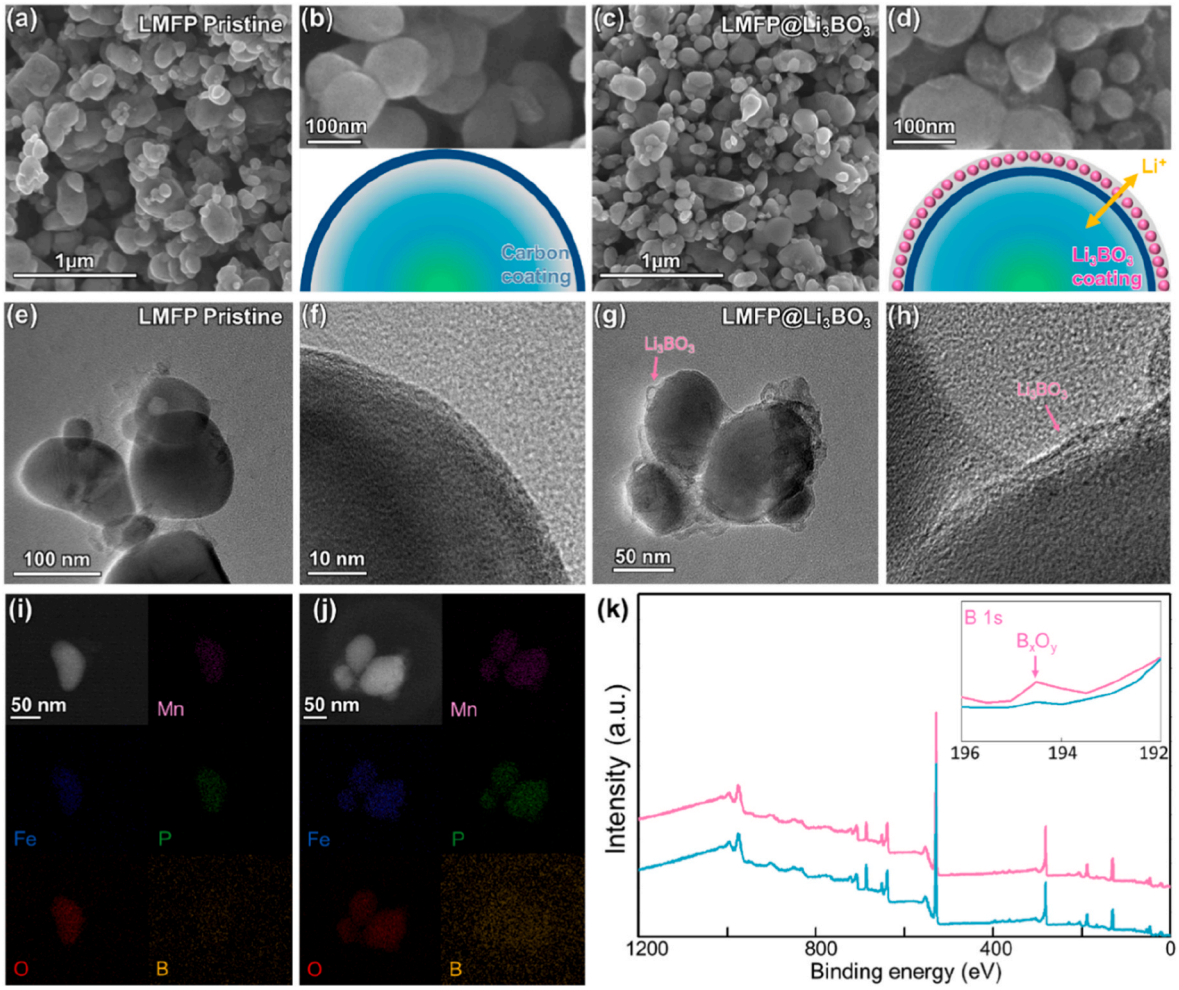


Fig. 2. Microstructural characterizations of as-prepared samples. (a) SEM images and (b) Schematic Diagram of LMFP pristine. (c) SEM images and (d) schematic illustration of LMFP@LBO particles. (e) TEM image and (f) HR-TEM image of LMFP pristine. (g) TEM image and (h) HR-TEM image of LMFP@LBO. EDX elemental mapping of (i) LMFP of (j) LMFP@LBO. (k) X-ray photoelectron spectroscopy spectra of LMFP pristine and LMFP@LBO.

modification process. Furthermore, the elemental mapping results show similar elemental distributions in both samples, implying that the overall elemental composition of the LMFP particles is minimally affected by the Li_3BO_3 coating. The carbon (C) and phosphorus (P) map appears weaker due to its low X-ray yield and additional attenuation of low-energy X-rays by the Li_3BO_3 coating layer. To gain deeper insights into the surface structures, transmission electron microscopy (TEM) analyses were performed. Fig. 2e and f shows representative TEM images of pristine LMFP, clearly revealing the carbon coating layer on the particle surfaces. For the coated sample, TEM images of LMFP@LBO are exhibited in Fig. 2g and h. Notably, the high-resolution image in Fig. 2h clearly visualizes a uniform amorphous coating layer with a thickness of approximately 3 nm along the surface of the LMFP particle, while the original carbon coating also remains well-preserved. This amorphous nature, characterized by the absence of regular lattice fringes, explains why the coating is not detectable by XRD. To verify the chemical composition of this surface layer, EDX elemental mapping was conducted, as shown in Fig. 2i and j. Both pristine and coated samples demonstrate homogeneous distributions of the primary elements (Mn, Fe, P, and O) within the particles. For the LMFP@LBO sample, despite the inherent difficulty in detecting light elements such as boron (B) with EDX, the B elemental map in Fig. 2j shows a faint signal distributed along the particle outline. This result indicates that the amorphous layer observed in TEM contains the intended boron precursor, confirming the successful formation of a functional amorphous lithium borate protective layer.

Additional TEM images and corresponding spectra of the LMFP@LBO sample are presented in Fig. S10–S11. To further confirm the presence of boron, X-ray photoelectron spectroscopy (XPS) analysis was conducted. The full XPS survey spectra shown in Fig. 2k clearly exhibit a distinct peak at approximately 194 eV associated with the B_xO_y bonding, present exclusively in the LMFP@LBO sample. Individual high-resolution XPS spectra of the O 1s, Li 1s, B 1s, Mn 2p, Fe 2p, and P 2p regions for the LMFP@LBO sample are provided in Fig. S12. Specifically, Fig. S12a shows a strong O 1s peak around 530 eV, corresponding to stable Metal–O bonds, and Fig. S12b displays a clear Li 1s peak at around 54 eV. The distinct B 1s peak at approximately 190 eV in Fig. S12c further confirms the formation of the Li_3BO_3 coating layer. The full XPS survey spectra shown in Fig. 2k clearly exhibit a distinct peak near 194 eV, corresponding to B_xO_y bonding, observed exclusively in the LMFP@LBO sample. High-resolution XPS spectra of O 1s, Li 1s, B 1s, Mn 2p, Fe 2p, and P 2p for LMFP@LBO are presented in Fig. S12. The O 1s spectrum displays a strong peak at approximately 530 eV, indicating the presence of stable metal–oxygen bonds, while the Li 1s spectrum shows a well-defined peak around 54 eV, confirming lithium incorporation. A pronounced peak around 190 eV in the B 1s spectrum provides further evidence for the successful formation of the Li_3BO_3 layer on the surface. In addition, the Mn 2p and Fe 2p spectra reveal clear doublet features with Mn peaks centered near 642 and 653 eV, and Fe peaks appearing around 711 and 724 eV. These results suggest that the chemical states of

Mn and Fe are well-preserved after the coating process. The P 2p spectrum shows a prominent peak near 133 eV, which is indicative of structurally stable phosphate groups, further confirming the chemical integrity of the LMFP framework. Collectively, these comprehensive microstructural and spectroscopic analyses provide strong evidence that the Li_3BO_3 coating effectively stabilizes the LMFP particle surface without significantly altering the original carbon coating or elemental distribution, ultimately enhancing the electrochemical stability and structural integrity of the LMFP cathode material.

Fig. 3a and b shows the X-ray diffraction (XRD) patterns of LMFP and LMFP@LBO. Both samples display the typical olivine structure, whereas the broad feature between 10° and 20° in LMFP@LBO (Fig. 3c) indicates the presence of an amorphous phase. The coating formed via the low-temperature (60 °C) solution process is amorphous or poorly crystalline, rendering its diffraction peaks undetectable by XRD. Nevertheless, EDX and XPS analyses confirm the coexistence of Li, B, and O in a ratio consistent with Li_3BO_3, verifying the formation of a Li–B–O bonding network on the particle surface. This type of amorphous coating is a characteristic feature of boron-based materials and may be attributed to the formation of a glassy Li_2O–B_2O_3 phase [26,45–47]. The Fourier Transform Infrared Spectroscopy (FT-IR) spectra of both samples (Fig. 3d) show no significant differences, implying that the bulk compositions remain unchanged after coating. However, Raman spectra (Fig. 3e and f) show distinct differences. In LMFP@LBO, clear peaks appear near 980–990 cm^{-1} and 1040–1050 cm^{-1}, corresponding to BO_4 tetrahedral and BO_3 planar triangular units, respectively, confirming the successful incorporation of boron. D-band (~1340 cm^{-1}) and G-band (~1580 cm^{-1}) features from the carbon coating are observed in both samples, indicating the carbon coating remained intact during synthesis. To evaluate the impact of Li_3BO_3 coating on the surface and porosity of LMFP, Brunauer-Emmett-Teller (BET) surface area and pore structure analyses were conducted. As shown in Fig. 3g, both samples display typical type IV nitrogen adsorption–desorption isotherms, indicating the presence of mesoporous structures. LMFP@LBO exhibited higher nitrogen adsorption over the entire pressure range, with a steep increase near $P/P_0 \approx 1.0$ (169.23 cc/g for LMFP@LBO vs. 129.20 cc/g for LMFP), indicating increased specific surface area and porosity due to Li_3BO_3 coating. The corresponding pore size distribution curves show dominant pores in the 12–16 nm range for both samples, consistent with mesoporous characteristics. Notably, LMFP@LBO displays a broader and more intense pore distribution, suggesting the formation of a hierarchical porous network that facilitates electrolyte penetration and lithium-ion transport. Furthermore, BET surface area analysis from Supporting Information Fig. S13 confirms that LMFP@LBO exhibits a slightly higher surface area (23.901 m^2/g) than LMFP (22.492 m^2/g), indicating that the coating enhances surface reactivity and porosity without compromising particle integrity. The thermal stability of the LMFP@LBO was assessed using thermogravimetric analysis (TGA) and Differential Scanning Calorimetry (DSC), as shown in Fig. 3h and i. Both samples exhibited minimal weight loss below 600 °C under air, confirming their excellent thermal stability. The residual mass at 600 °C was approximately 98.92 % for LMFP and 98.82 % for LMFP@LBO, indicating negligible thermal decomposition. The DSC curve of LMFP shows a distinct endothermic peak above 400 °C, likely attributed to phase transitions or surface oxidation reactions. In contrast, LMFP@LBO exhibits a more moderate heat flow, suggesting that the LBO coating contributes to thermal stability by acting as a heat-buffering layer or suppressing side reactions at elevated temperatures.

3.3. Electrochemical characterization

To evaluate the electrochemical characteristics of the electrodes,

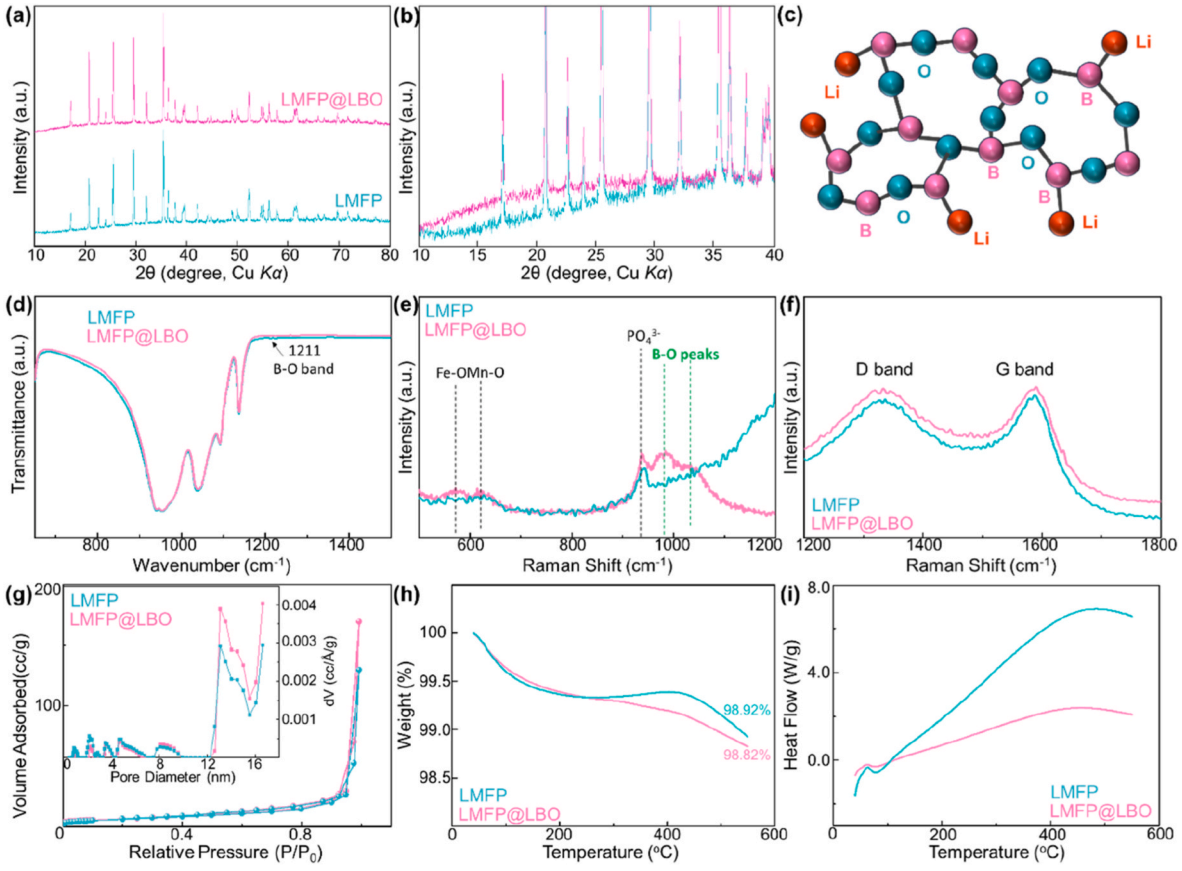


Fig. 3. (a), (b) XRD patterns of LMFP@LBO and LMFP. (c) XRD patterns of Li_3BO_3. (d) FTIR spectra of LMFP and LMFP@LBO. (d) FTIR profile and (e), (f) Raman spectra of LMFP and LMFP@LBO. (g) BET profile of LMFP and LMFP@LBO. (h) TGA curves and (i) DSC curves of LMFP and LMFP@LBO.

cyclic voltammetry (CV), galvanostatic charge-discharge (GCD), and electrochemical impedance spectroscopy (EIS) analyses were conducted (Fig. 4). Detailed experimental procedures, including electrochemical measurements and cell assembly, are provided in the Experimental Section.

Fig. 4a presents the CV curves of LMFP and LMFP@LBO electrodes at a scan rate of 0.5 mV/s. Both electrodes display two well-defined redox couples corresponding to Fe^{2+}/Fe^{3+} and Mn^{2+}/Mn^{3+} reactions, centered around 3.3–3.6 V and 4.0–4.2 V, respectively. Compared to pristine LMFP, the LMFP@LBO electrode exhibits higher current response and more symmetric peak shapes, indicating improved reversibility and enhanced charge-transfer kinetics enabled by the Li_3BO_3 surface layer. In the Fe redox couple, the LMFP@LBO electrode shows peak potentials of 3.52 V for oxidation and 3.35 V for reduction, with a voltage gap of approximately 170 mV. In contrast, pristine LMFP presents a broader gap of 210 mV, with oxidation and reduction peaks at 3.50 V and 3.29 V. Similarly, for the Mn redox couple, LMFP@LBO shows a narrower potential difference of 110 mV (4.16 V and 4.05 V) compared to 130 mV for LMFP (4.13 V and 4.00 V).

Fig. 4b illustrates the GCD profiles at various current densities. At 0.02 A/g, both electrodes deliver nearly identical discharge capacities,145.40 mAh/g for LMFP@LBO and 145.20 mAh/g for pristine LMFP. However, at 0.64 A/g, LMFP@LBO retains a significantly higher capacity of 94.04 mAh/g, while LMFP declines to 58.51 mAh/g. This enhanced rate capability reflects the effectiveness of the LBO coating in facilitating Li^+ transport. Additional charge-discharge curves obtained at intermediate current densities are provided in Fig. S14 and S15, which further confirm the superior capacity retention of LMFP@LBO under high-rate conditions.

The rate performance shown in Fig. 4c further highlights the kinetic

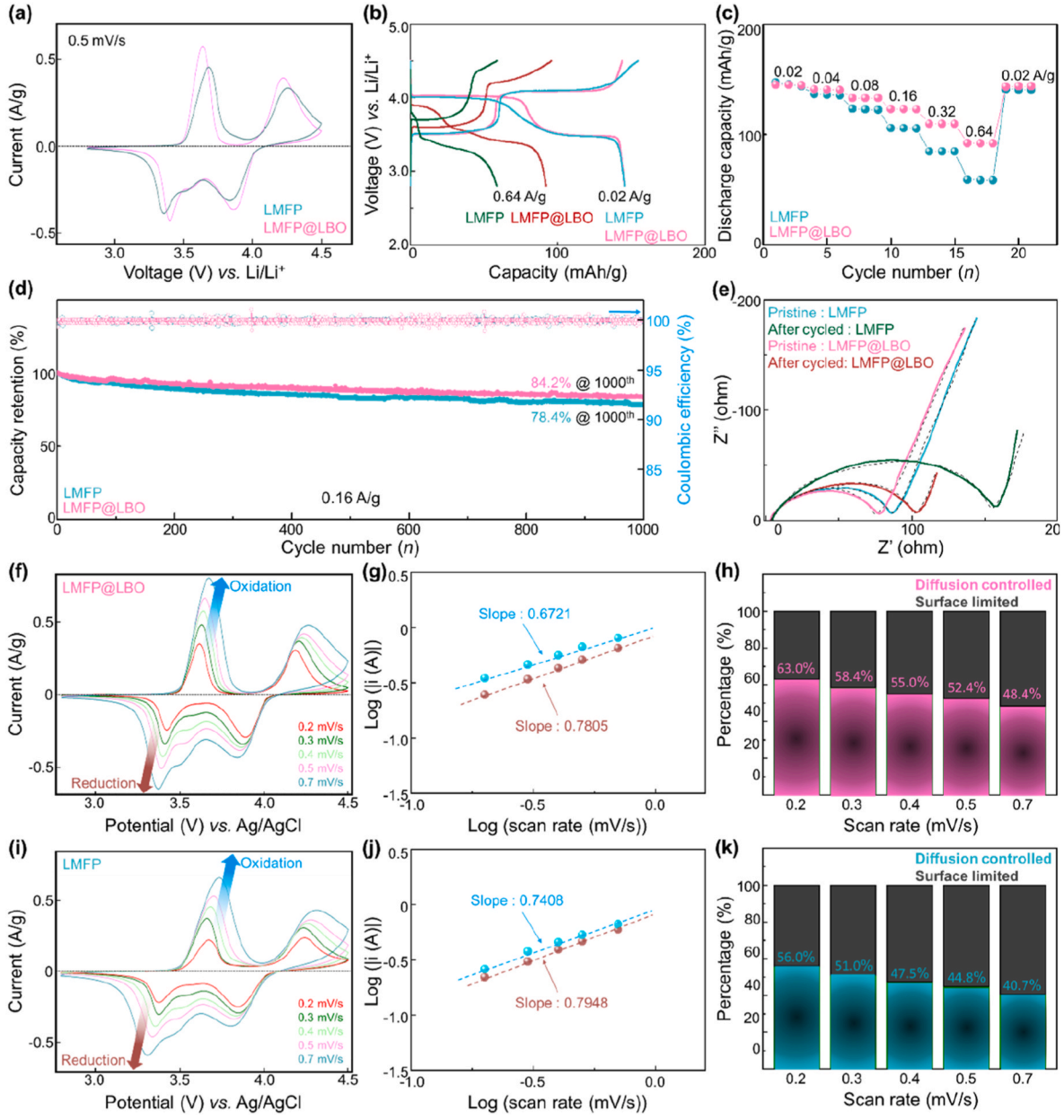


Fig. 4. Comparative electrochemical performances of LMFP and LMFP@LBO. (a) CV curves at a scan rate of 0.5 mV/s. (b) Galvanic charge/discharge profiles at current density 0.02 A/g and 0.64 A/g. (c) Rate cycling curves. (d) Cycling performance. (e) Nyquist Plot in EIS curves. CV cycle measured at scan rates from 0.2 to 0.7 mV/s for (f) LMFP@LBO and (i) LMFP. calculation of b-values based on the relationship between specific cathodic peak current and scan rate for (g) LMFP@LBO and (j) LMFP. calculation of surface-limited and diffusion-controlled ion reaction ratios for (h) LMFP@LBO and (k) LMFP.

advantage of the LBO-coated electrode. Even after increasing the current density from 0.02 A/g to 0.64 A/g and returning to 0.02 A/g, LMFP@LBO maintains a capacity of 91.85 mAh/g, significantly outperforming LMFP, which retains only 58.51 mAh/g. This behavior is attributed to the amorphous nature of the Li_3BO_3 coating, which not only provides ionic conductivity but also creates favorable pathways for lithium-ion transport [48–51].

Cycling performance over 1000 cycles at 0.16 A/g is shown in Fig. 4d. LMFP@LBO retains 84.2 % of its initial capacity, whereas LMFP retains only 78.4 %. The improved durability is primarily attributed to the suppression of Mn dissolution and stabilization of the electrode structure by the LBO surface layer. Both samples exhibit Coulombic efficiencies close to 100 %, indicating excellent cycling reversibility. To provide direct evidence for the suppression of Mn dissolution, inductively coupled plasma–optical emission spectrometry (ICP–OES) was conducted on the electrolytes collected from cells after 1000 cycles (Table S1). The electrolyte from the pristine LMFP cell exhibited a much higher Mn concentration, whereas the LMFP@LBO cell showed a substantially lower value, confirming that the Li_3BO_3 coating effectively suppresses Mn dissolution. This quantitative result supports that the enhanced cycling stability of LMFP@LBO originates from the protection of the active surface against electrolyte corrosion. Furthermore, the lithium-rich amorphous Li_3BO_3 coating may also act as a lithium reservoir, compensating for the irreversible lithium loss that typically occurs during SEI formation in the initial cycles. This lithium replenishment effect, combined with the suppression of Mn dissolution and enhanced interfacial stability, contributes to the excellent long-term cycling stability of the LMFP@LBO cathode. Fig. 4e compares the EIS results before and after cycling. Initially, the charge-transfer resistance of LMFP@LBO is lower (79.8 Ω) than that of LMFP (88.3 Ω). After 1000 cycles, LMFP@LBO maintains a relatively small increase in resistance (106 Ω), whereas LMFP exhibits a much larger increase (159 Ω). These results highlight the role of the LBO coating in enhancing interfacial conductivity and mitigating resistance buildup during long-term operation. Fig. 4f and i displays CV curves of both electrodes measured at scan rates ranging from 0.2 to 0.7 mV/s. In both cases, peak current increases with scan rate, as expected. However, LMFP@LBO consistently shows stronger and sharper redox peaks, confirming that the surface coating facilitates faster kinetics and greater electrochemical stability. To further understand the reaction mechanisms, log(i)–log(v) plots were constructed (Fig. 4g and j). The slope values for LMFP@LBO are 0.6721 for oxidation and 0.7805 for reduction, while LMFP exhibits slightly higher values of 0.7408 and 0.7948. These values suggest that LMFP@LBO achieves a better balance between diffusion-controlled and surface-limited processes, contributing to more efficient lithium-ion storage. Fig. 4h and k quantify the relative contributions of diffusion-controlled and surface-limited processes. At the highest scan rate of 0.7 mV/s, the surface-limited contribution is calculated to be 48.4 % for LMFP@LBO and 40.7 % for LMFP. This implies that LMFP@LBO sustains a higher proportion of diffusion-controlled behavior across all scan rates, affirming the superior lithium transport enabled by the Li_3BO_3 interface. The reduced interfacial resistance and improved ionic pathways contribute to the observed performance benefits. Further insight is provided in Supporting Fig. S16 and S17. Linear fitting of the peak current versus square root of scan rate in Fig. S16a yields slope values of 0.4753 (oxidation) and 0.5905 (reduction) for LMFP@LBO, indicating a dominant diffusion-controlled mechanism. In comparison, pristine LMFP (Fig. S17a) shows slope values of 0.5596 and 0.5155, suggesting slightly more surface-limited behavior. Bar charts in Fig. S16b and c and, S7b,c show the quantitative breakdown of capacitive and diffusion-controlled contributions across different scan rates. At 0.7 mV/s, LMFP@LBO achieves diffusion-controlled contributions of 62.6 % in the oxidation process and 37.4 % in the reduction process, whereas LMFP records 46.7 % and 34.8 %, respectively. These results demonstrate that Li_3BO_3 coating facilitates more efficient Li^+ diffusion and ensures greater stability under high-rate operation. Fig. S16 and S17 provide further insights into the lithium-ion storage mechanism of LMFP and LMFP@LBO electrodes by evaluating their capacitive and diffusion-controlled contributions. In Fig. S16a, LMFP@LBO exhibits a lower slope (0.4753 for oxidation, 0.5905 for reduction) compared to pristine LMFP (Fig. S17a: 0.5596 for oxidation, 0.5155 for reduction), indicating a relatively greater influence of diffusion-controlled processes. Additionally, the quantitative analysis in Figs. S16b,c and S17b,c shows that at all scan rates (0.2–0.7 mV/s), LMFP@LBO consistently demonstrates higher diffusion-controlled contributions than LMFP. For instance, at 0.7 mV/s, the diffusion-controlled ratio reaches 62.6 % (oxidation) and 37.4 % (reduction) for LMFP@LBO, while LMFP exhibits 46.7 % and 34.8 %, respectively. In summary, surface engineering via Li_3BO_3 coating effectively shifts the electrochemical mechanism of LMFP cathodes toward diffusion-controlled behavior, thereby enhancing both lithium storage capacity and high-rate performance.

To further quantify the lithium-ion diffusion kinetics, Galvanostatic Intermittent Titration Technique (GITT) measurements were performed (Fig. S18). While the calculated diffusion coefficients (D_{Li+}) derived from the GITT profiles were within a similar order of magnitude for both electrodes, a distinct difference was evident in the average values. The pristine LMFP electrode exhibited an average D_{Li+} of 7.58×10^{-9} cm^2/s, whereas the LMFP@LBO electrode delivered a nearly twofold higher value of 1.41×10^{-8} cm^2/s. This result provides direct evidence that the LBO coating layer significantly enhances the lithium-ion transport kinetics, which is consistent with the superior rate capability observed in Fig. 4c and the lower charge-transfer resistance (R_{ct}) shown in Fig. 4e.

Collectively, these results demonstrate that the Li_3BO_3 surface coating substantially improves the electrochemical performance of the LMFP cathode. While the initial discharge capacity remained largely unaffected, significant enhancements were observed in the rate capability and long-term cycling stability. Specifically, at a high current density of 0.64 A/g, the LBO-coated electrode exhibited a discharge capacity of 91.85 mAh/g, a 57 % improvement over the 58.51 mAh/g retained by the pristine LMFP. Furthermore, in terms of long-term durability, the capacity decay after 1000 cycles was suppressed by 27 %, with the coated electrode retaining 84.2 % of its initial capacity compared to 78.4 % for the pristine counterpart. This superior performance is directly correlated with the enhanced interfacial kinetics, as substantiated by electrochemical impedance spectroscopy. The Li_3BO_3 coating not only reduced the initial charge-transfer resistance (Rct) from 88.3 Ω to 79.8 Ω (a 9.6 % decrease) but also mitigated its subsequent increase during cycling. The change in Rct after 1000 cycles for the coated electrode was merely 26.2 Ω (from 79.8 to 106 Ω), representing a 63 % suppression of the resistance growth observed in the pristine electrode, which exhibited an increase of 70.7 Ω (from 88.3 to 159 Ω). These findings conclusively establish that the Li_3BO_3 surface modification is a highly effective strategy for stabilizing the electrode–electrolyte interface, thereby enhancing the rate performance and cycling stability of LMFP cathodes for practical high-performance lithium-ion batteries.

3.4. LMFP@LBO stability analysis

To evaluate the structural integrity and surface chemistry of the electrodes after cycling, a comprehensive set of characterizations including SEM, TEM, HRTEM, EDX mapping, XRD, FT-IR, Raman spectroscopy, and XPS analyses were conducted, as shown in Fig. 5. The SEM images in Fig. 5a–d illustrate the morphological evolution of pristine LMFP and LMFP@LBO electrodes before and after cycling. The cycled LMFP electrode exhibits clear evidence of particle fracture and aggregation, likely due to weakened interparticle adhesion. In contrast, the LMFP@LBO electrode retains uniform particle morphology with well-defined boundaries, indicating that the Li_3BO_3 coating effectively suppresses structural collapse during long-term operation. High-resolution TEM analyses further reveal the presence of a conformal amorphous LBO layer with a uniform thickness of approximately 5 nm on the surface of LMFP@LBO particles, as shown in Fig. 5e. The

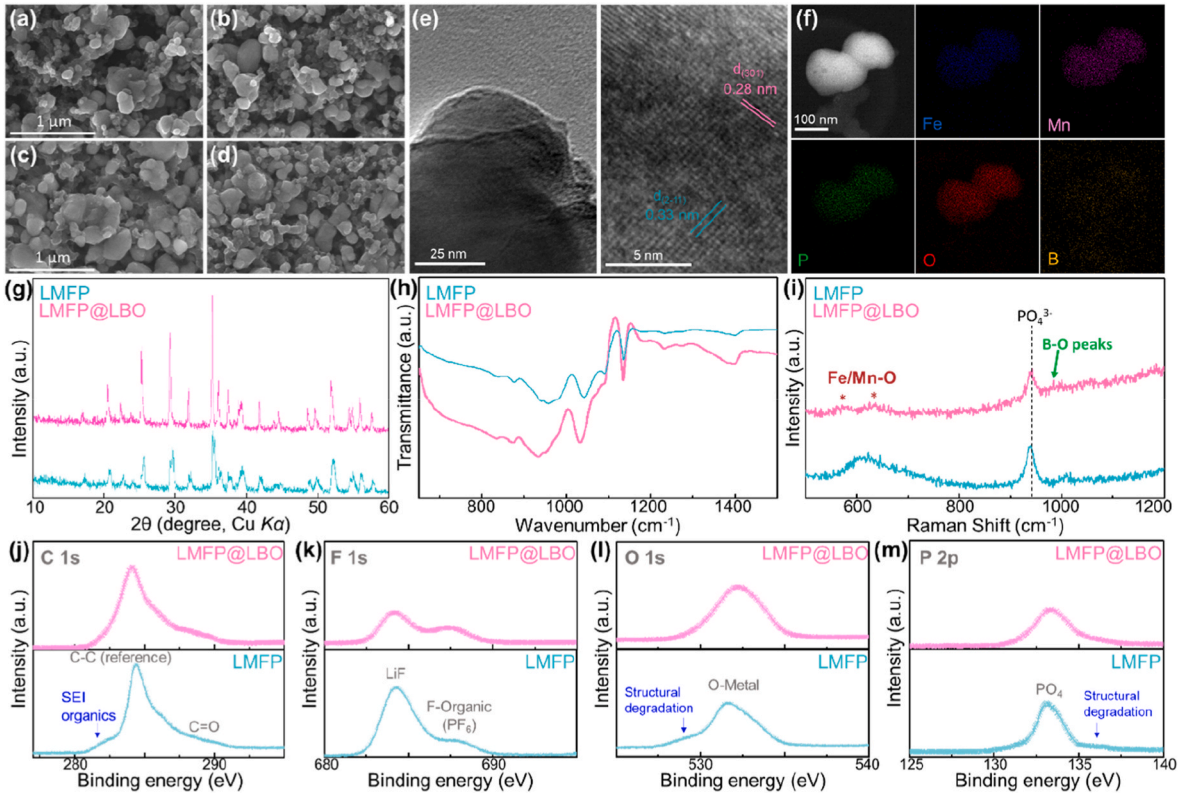


Fig. 5. Structural and surface properties of LMFP and LMFP@LBO electrodes after cycling. SEM images of (a) LMFP and (b) LMFP@LBO electrodes before cycling, and (c) LMFP and (d) LMFP@LBO electrodes after cycling. (e) TEM and HRTEM images of LMFP@LBO. (f) TEM-EDX elemental mapping of LMFP@LBO particles. (g) Comparative XRD patterns of cycled LMFP and LMFP@LBO electrodes. (h) Comparative FT-IR spectra of LMFP and LMFP@LBO after cycling. (i) Raman spectroscopy spectra highlighting Fe/Mn–O, PO_4^{3-}, and characteristic B–O peaks in cycled electrodes. (j–m) XPS spectra for detailed surface chemistry evaluation of cycled LMFP and LMFP@LBO electrodes (j) C 1s region, (k) F 1s region, (l) O 1s region, and (m) P 2p region, illustrating differences in surface chemical states and interfacial stability after cycling.

observed lattice fringes of 0.28 nm and 0.33 nm correspond well to the (301) and (2−11) planes of the LMFP crystal, confirming the preservation of the bulk crystal structure. Elemental mapping results in Fig. 5f show that Fe, Mn, P, and O are uniformly distributed throughout the LMFP@LBO particles, while B is clearly concentrated at the surface. This indicates that the Li_3BO_3 coating remains intact even after extensive cycling, continuing to act as a protective surface layer and contributing to the structural stability of the electrode. XRD patterns in Fig. 5g further support this conclusion. LMFP@LBO retains sharp and well-defined diffraction peaks after cycling, whereas pristine LMFP shows peak broadening and reduced intensity, suggesting partial loss of crystallinity. This demonstrates the role of the Li_3BO_3 layer in maintaining structural integrity during repeated lithium insertion and extraction. FT-IR spectra shown in Fig. 5h reveal that LMFP@LBO maintains all characteristic absorption bands with little change compared to pristine LMFP, suggesting that the internal structure remains stable without significant decomposition. Raman spectroscopy results in Fig. 5i also show distinct peaks corresponding to PO_4^{3-} and B–O bonding in the LMFP@LBO electrode. The Fe–O and Mn–O vibration modes are clearly retained, indicating that the surface modification with Li_3BO_3 effectively mitigates structural damage. In contrast, the pristine LMFP shows merging of Fe–O and Mn–O peaks, likely due to Mn dissolution and collapse of the transition-metal lattice structure.

To further examine the surface chemical state after cycling, XPS analysis was conducted, as shown in Fig. 5j–m. The C 1s spectrum of the LMFP electrode reveals prominent peaks corresponding to C–O and C=O species, indicative of SEI formation caused by electrolyte decomposition. These features are largely suppressed in LMFP@LBO, confirming that the Li_3BO_3 layer effectively inhibits side reactions at the electrode–electrolyte interface. Similarly, the F 1s spectrum of LMFP exhibits strong signals for LiF and organic fluorine compounds, indicating significant electrolyte degradation. These peaks are barely detectable in LMFP@LBO, suggesting that the Li_3BO_3 coating prevents the formation of such decomposition products. The O 1s spectrum of LMFP shows an additional shoulder associated with structural degradation, while LMFP@LBO maintains a single, well-defined peak corresponding to stable metal–oxygen bonds, confirming minimal structural damage. Finally, in the P 2p spectrum, LMFP displays secondary peaks indicative of phosphate decomposition. In contrast, LMFP@LBO exhibits only a single, sharp PO_4^{3-} peak, confirming the protective effect of the Li_3BO_3 layer on the integrity of the phosphate framework. These results collectively demonstrate that the Li_3BO_3 surface coating plays a critical role in enhancing the structural and interfacial stability of LMFP cathodes during long-term cycling.

4. Conclusions

In this study, we successfully developed an LMFP cathode material with significantly enhanced electrochemical performance and long-term cycling stability through the introduction of a lithium-rich Li_3BO_3 coating layer. Utilizing soft-BV computational analysis, we systematically identified Li_3BO_3 as an optimal coating candidate, exhibiting a low Li^+ diffusion barrier of approximately 0.321 eV, ideal for promoting efficient lithium-ion conduction within the electrode structure. Structural visualizations further confirmed a favorable lithium diffusion pathway along both *b*-axis (0.396 eV) and *c*-axis (0.321 eV) directions within Li_3BO_3.

Comprehensive electrochemical and physicochemical characterizations clearly demonstrated the effectiveness of Li_3BO_3 coating in overcoming intrinsic limitations of LMFP. Specifically, LMFP@LBO showed

significantly improved rate capability, maintaining a high discharge capacity of approximately 91.85 mAh/g at a high current density of 0.64 A/g, while pristine LMFP retained only 58.51 mAh/g under the same conditions. Moreover, long-term cycling tests at 0.16 A/g demonstrated superior cycling stability for LMFP@LBO, exhibiting remarkable capacity retention of 84.2 % after 1000 cycles, compared to 78.4 % for pristine LMFP. These improvements result from the protective effect of the amorphous Li_3BO_3 layer, which effectively reduces interfacial resistance and suppresses Mn dissolution arising from Jahn–Teller distortion. Overall, this work effectively demonstrates that employing an amorphous lithium-rich Li_3BO_3 surface coating significantly enhances both electrochemical performance and structural durability of LMFP cathodes. The surface modification strategy proposed in this study offers a highly feasible and scalable approach to designing next-generation LIBs cathode materials with low cost and high stability and is expected to play a key role in the development of high-energy-density and long-cycle-life rechargeable batteries.

CRediT authorship contribution statement

Seunghyeop Baek: Writing – original draft, Formal analysis, Data curation. **Dedy Setiawan:** Writing – review & editing, Writing – original draft. **Hyeongseok Lee:** Formal analysis. **Byung Gon Kim:** Supervision, Resources. **Munseok S. Chae:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jpowsour.2025.238865.

Data availability

Data will be made available on request.

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